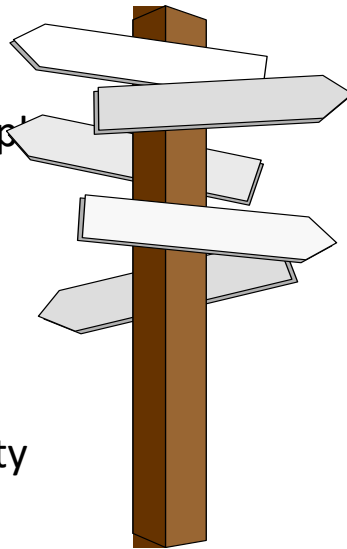


Design Metrics

Major Design Challenges

- Microscopic issues

- ultra-high speeds
- power dissipation and supply rail drop
- growing importance of interconnect
- noise, crosstalk
- reliability, manufacturability
- clock distribution



- Macroscopic issues

- time-to-market
- design complexity (millions of gates)
- high levels of abstractions
- design for test
- reuse and IP, portability
- tool interoperability

Year	Tech.	Complexity	Frequency	3 Yr. Design Staff Size	Staff Costs
1997	0.35	13 M Tr.	400 MHz	210	\$90 M
1998	0.25	20 M Tr.	500 MHz	270	\$120 M
1999	0.18	32 M Tr.	600 MHz	360	\$160 M
2002	0.13	130 M Tr.	800 MHz	800	\$360 M

Overview

- Digital integrated circuits experience exponential growth in complexity (Moore's law) and performance
- Design in the deep submicron (DSM) era creates new challenges
 - Devices become somewhat different, and hence new models are required for accurate simulation.
 - Global clocking becomes more challenging
 - Interconnect effects play a more significant role
 - Power dissipation may be *the* limiting factor

Design Issue of SOC

- Due to the use of various hard, firm, and soft cores from multiple vendors, the SoC design may contain a very high level of integration complexity, interfacing and synchronization issues, data management issues, design verification, and test, architectural, and system-level issues.
- The use of a wide variety of logic, memory, and analog/mixed-signal cores from different vendors can cause a wide range of problems in the design of SoC.

Cont...

- Portability Methodology-related Issues
 - Non-netlisted cores
 - Layout-dependent step sizes
 - Aspect ratio misfits
 - Hand-crafted layout
- Timing Issues
 - Clock redistribution
 - Hard core width and spacing disparities
 - Antenna rules mismatch
 - Timing reverification

Fundamental Design Metrics

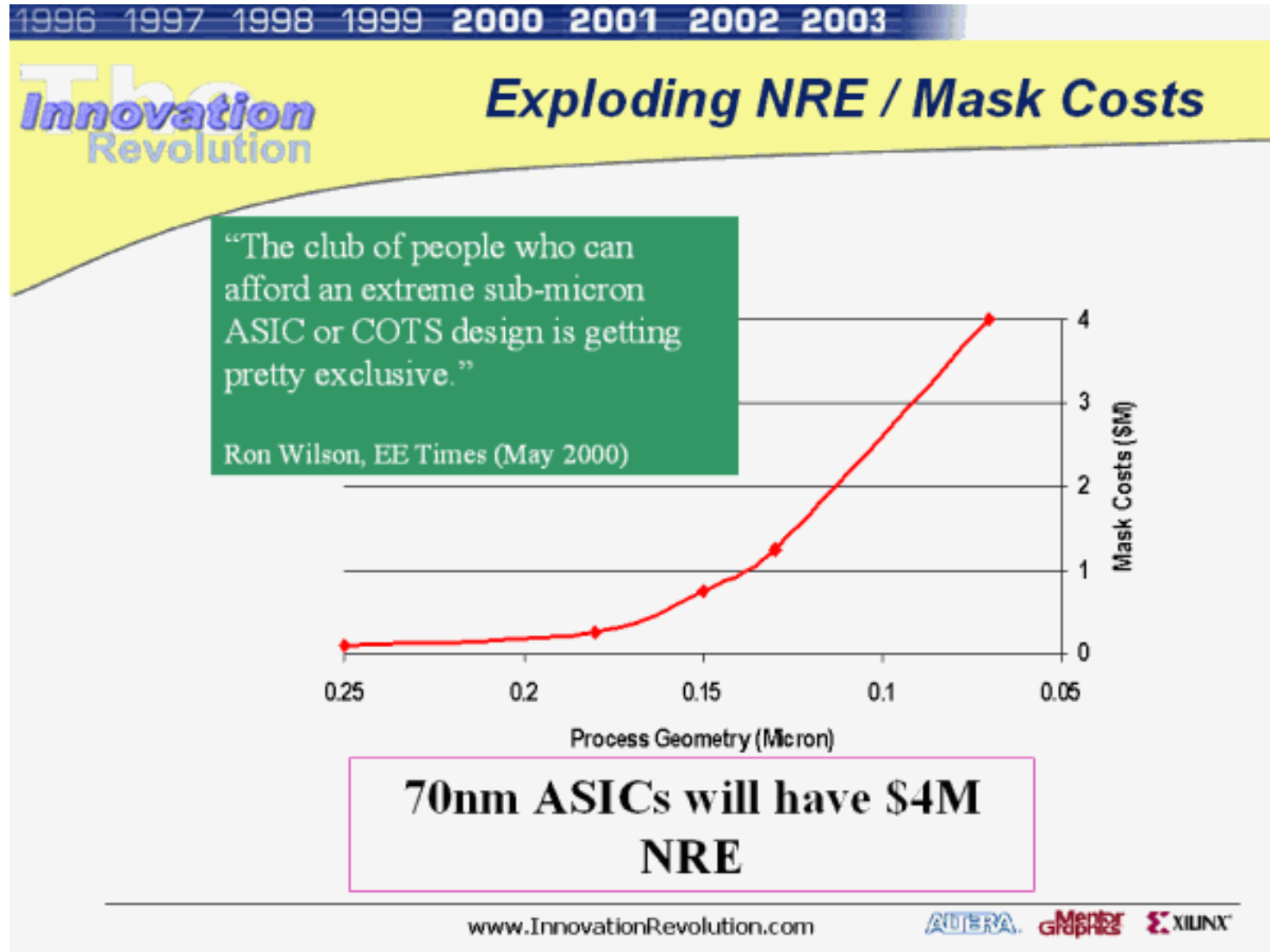
- Functionality
- Cost
 - NRE (fixed) costs - design effort
 - RE (variable) costs - cost of parts, assembly, test
- Reliability, robustness
 - Noise margins
 - Noise immunity
- Performance
 - Speed (delay)
 - Power consumption; energy
- Time-to-market
- Reusability

Cost of Integrated Circuits

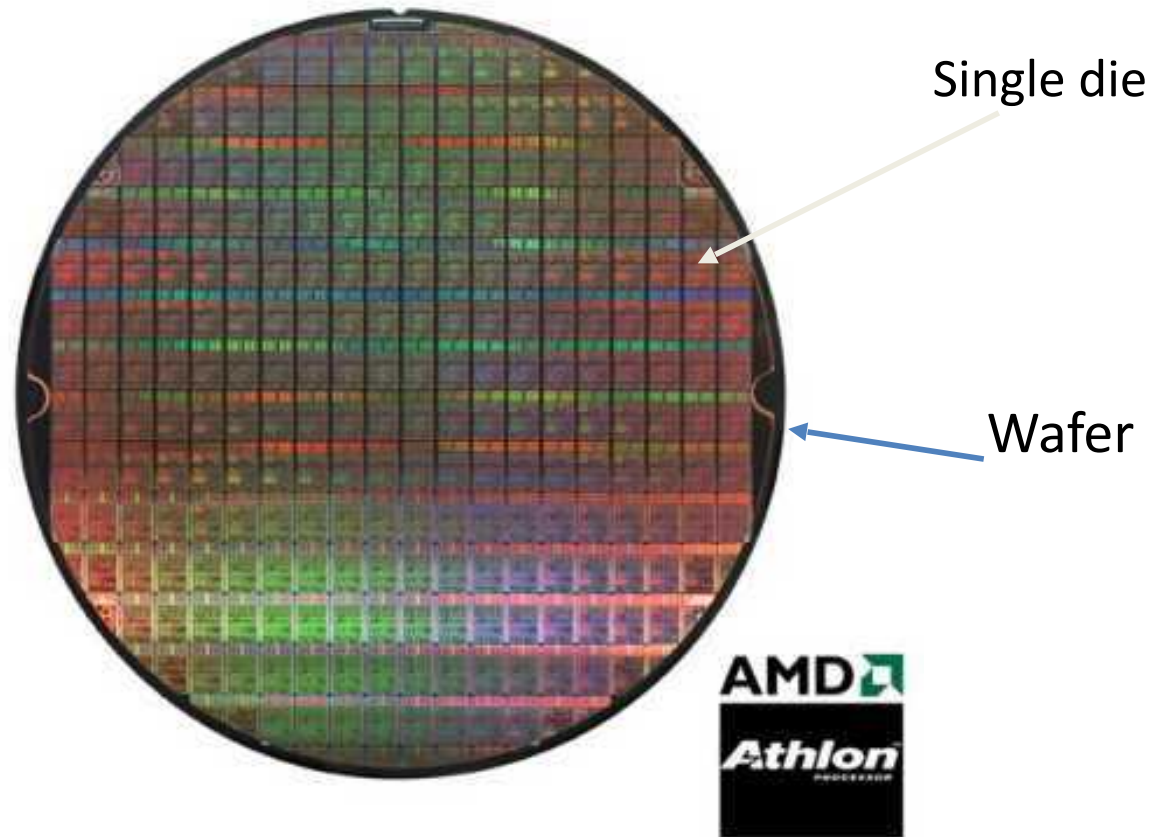
- NRE (non-recurring engineering) costs
 - Fixed cost to produce the design
 - design effort
 - design verification effort
 - mask generation
 - Influenced by the design complexity and designer productivity
 - More pronounced for small volume products
- Recurring costs – proportional to product volume
 - silicon processing
 - also proportional to chip area
 - assembly (packaging)
 - test

$$\text{cost per IC} = \text{variable cost per IC} + \frac{\text{fixed cost}}{\text{volume}}$$

NRE Cost is Increasing



Silicon Wafer

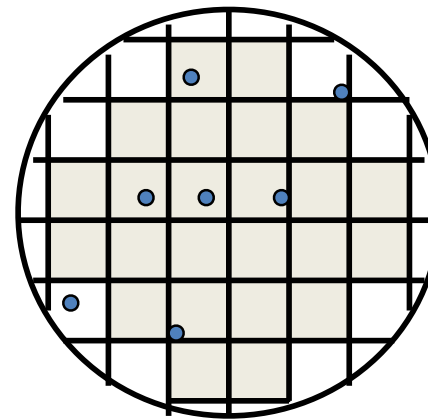
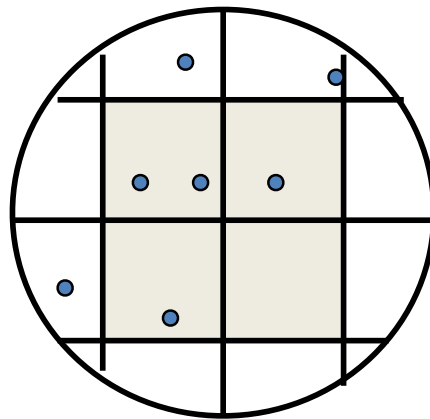


From <http://www.amd.com>

Recurring Costs

$$\text{cost of die} = \frac{\text{cost of wafer}}{\text{dies per wafer} \times \text{die yield}}$$

$$\text{dies per wafer} = \frac{\pi \times (\text{wafer diameter}/2)^2}{\text{die area}} - \frac{\pi \times \text{wafer diameter}}{\sqrt{2} \times \text{die area}}$$



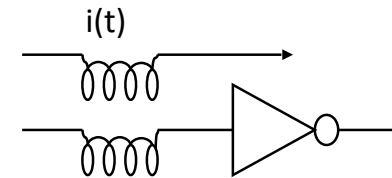
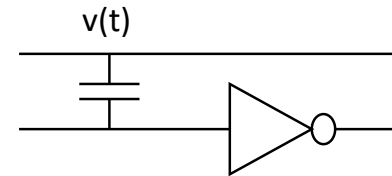
Reliability

Noise in Digital Integrated Circuits

- **Noise** – unwanted variations of voltages and currents at the logic nodes

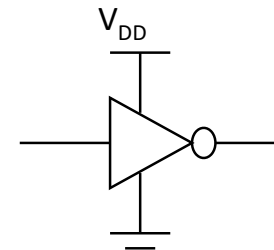
□ from two wires placed side by side

- capacitive coupling
 - voltage change on one wire can influence signal on the neighboring wire
 - cross talk
- inductive coupling
 - current change on one wire can influence signal on the neighboring wire



□ from noise on the power and ground supply rails

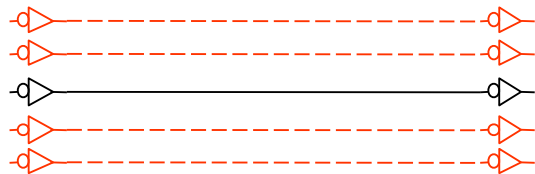
- can influence signal levels in the gate



Example of Capacitive Coupling

- Signal wire glitches as large as 80% of the supply voltage will be common due to crosstalk between neighboring wires as feature sizes continue to scale

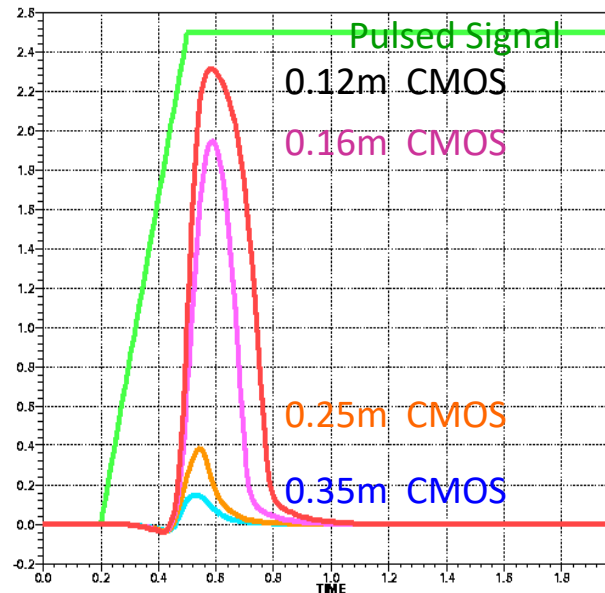
Crosstalk vs. Technology



Black line quiet

Red lines pulsed

Glitches strength vs technology

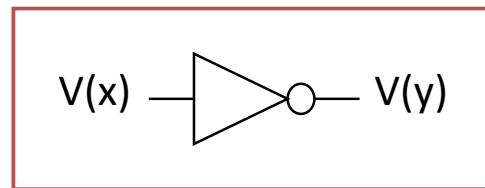


From Dunlop, Lucent, 2000

Static Gate Behavior

- Steady-state parameters of a gate – *static behavior* – tell how robust a circuit is with respect to both variations in the manufacturing process and to noise disturbances.
- Digital circuits perform operations on Boolean variables $x \in \{0,1\}$
- A logical variable is associated with a *nominal voltage level* for each logic state

$$1 \Leftrightarrow V_{OH} \text{ and } 0 \Leftrightarrow V_{OL}$$



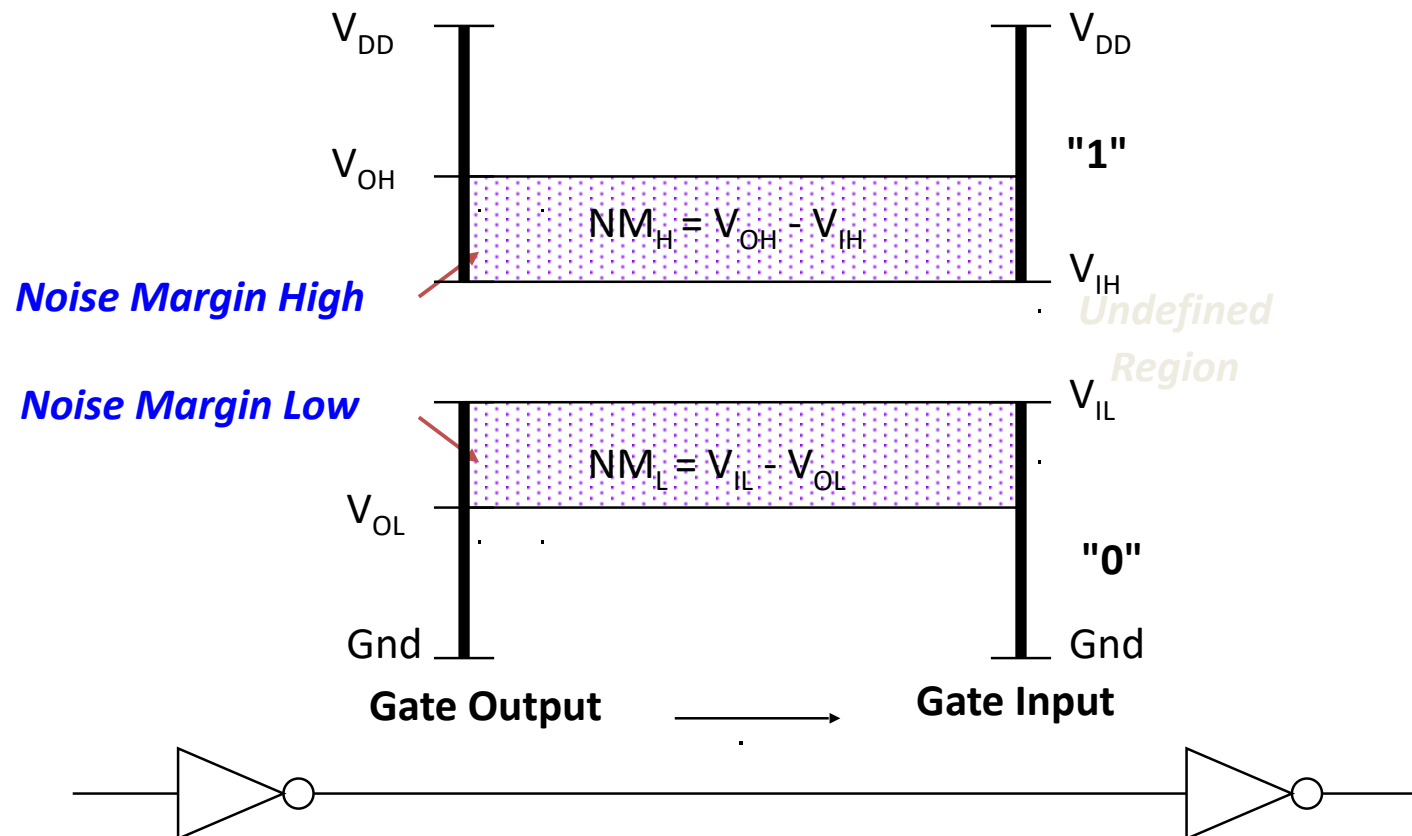
$$V_{OH} = ! (V_{OL})$$

$$V_{OL} = ! (V_{OH})$$

- Difference between V_{OH} and V_{OL} is the logic or *signal swing* V_{sw}

Noise Margins

- For robust circuits, want the “0” and “1” intervals to be as large as possible



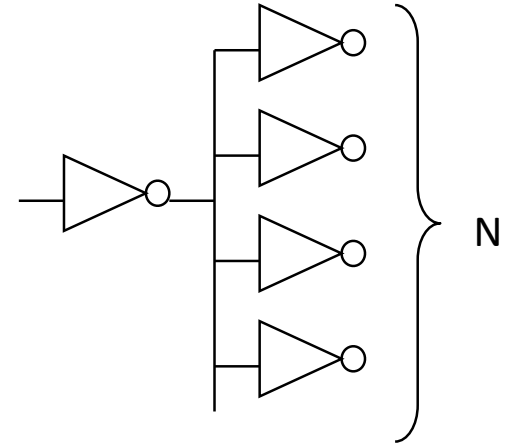
- Large noise margins are desirable, but not sufficient ...

Noise Immunity

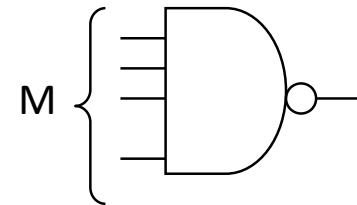
- ❑ Noise margin expresses the ability of a circuit to overpower a noise source
 - noise sources: supply noise, cross talk, interference, offset
- ❑ Absolute noise margin values are deceptive
 - a floating node is more easily disturbed than a node driven by a low impedance (in terms of voltage)
- *Noise immunity* expresses the ability of the system to process and transmit information correctly in the presence of noise
- For good noise immunity, the signal swing (i.e., the difference between V_{OH} and V_{OL}) and the noise margin have to be large enough to overpower the impact of fixed sources of noise

Fan-In and Fan-Out

- Fan-out – number of load gates connected to the output of the driving gate
 - gates with large fan-out are slower



- Fan-in – the number of inputs to the gate
 - gates with large fan-in are bigger and slower



Time To Market

- Time To market for SOC Design should be appropriate.
- Design reusability contributes to reducing Time To Market for SOC Design.

Design Deliverables : Soft IP Core

- Synthesizable Verilog/VHDL
- Example synthesis script
- RTL compiled module
- Structural compiled module
- Design, timing, and synthesis shells
- Functional simulation testbench
- Installation script
- Bus functional models and monitors used in testbenches
- Testbenches with sample verification tests
- Cycle-based simulation or emulation models
- Bus functional models
- Application note that describes signal slew rate at the inputs, clock skew tolerance, output-loading range, and test methodology

Design Deliverables : Hard IP Core

- Installation scripts
- ISA or behavioral model of the core
- Bus functional and fully functional models for the core
- Cycle-based emulation model (on request)
- Floor planning, timing, and synthesis models
- Functional simulation testbench
- Bus functional models and monitors used in testbenches
- Testbenches with verification tests
- Manufacturing tests
- GDSII with technology file